Physics 4220H: Assignment III

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Problem 1. Starting from the following expression for (the square of) the complex propagation constant in a conductor,

$$\widetilde{k}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega = \mu\omega\left[\epsilon\omega + i\sigma\right]$$

Go through the algebra to find expressions for the real ($\|$) and imaginary (κ) parts of the propagation constant. i.e. find $\|$ and κ, where $\widetilde{k} = \| + \kappa$.

Solution 1. To do this I will write $\widetilde{k}$ in polar form, apply the square root, then use the half-angle formulae

$$\cos(x/2) = \sqrt{\frac{1+\cos(x)}{2}} \quad \sin(x/2) = \sqrt{\frac{1-\cos(x)}{2}}$$

and the formulae (derivable by considering a triangle with sides 1, x and $\sqrt{1+x^2}$)

$$\cos(\arctan(x)) = \frac{1}{\sqrt{1+x^2}} \quad \sin(\arctan(x)) = \frac{x}{\sqrt{1+x^2}}.$$

To start, we can write

$$\widetilde{k}^2 = Ke^{i\phi}$$

with

$$K = \left|\widetilde{k}\right| = \mu\omega\sqrt{(\epsilon\omega)^2+\sigma^2} \quad \phi = \arctan\Big(\frac{\sigma}{\epsilon\omega}\Big).$$

We then write

$$\widetilde{k} = \sqrt{Ke^{i\phi}} = \sqrt{K}e^{i\phi/2} = \sqrt{\mu\omega\sqrt{(\epsilon\omega)^2+\sigma^2}}\left[\cos\left(\frac{1}{2}\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\right) + i\sin\left(\frac{1}{2}\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\right)\right]$$

$$= \omega\sqrt{\mu\epsilon}\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}}\left[\cos\left(\frac{1}{2}\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\right) + i\sin\left(\frac{1}{2}\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\right)\right]$$

and then use the half-angle identities to write this as

$$\widetilde{k} = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}}\left[\sqrt{1+\cos\Big(\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\Big)} + i\sqrt{1-\cos\Big(\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\Big)}\right]$$

and then apply the $\cos(\arctan(x))$ identity to rewrite that as

$$\widetilde{k} = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}}\left[\sqrt{1+\frac{1}{\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}}} + i\sqrt{1-\frac{1}{\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}}}\right].$$

Wow! Now multiply in the square root out front,

$$\widetilde{k} = \omega\sqrt{\frac{\mu\epsilon}{2}}\left[\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}+1} + i\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}-1}\right].$$

so

$$\| = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}+1}$$

and

$$\kappa = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}-1}.$$

Problem 2. Household microwave ovens operate at a frequency of $\nu = 2.54\,\mathrm{GHz}$. Imagine that you leave a $20\,^\circ\mathrm{C}$ stainless-steel spoon ($s = 1.5\times10^6\,\mathrm{S\,m^{-1}}$, $\mu \approx \mu_0$, $\epsilon \approx \epsilon_0$) next to a bowl of soup in the microwave.

(a) Estimate the proportion of the incident field that is reflected and transmitted from/into the spoon.

(b) Estimate the skin-depth over which the microwaves that enter the handle are absorbed

(c) Imagine that the full power of a 1 kW microwave is incident onto the spoon's handle, after which any reflected power gets absorbed by the soup. If the spoon weighs 30 g and the steel has a heat capacity of $450\,\mathrm{J\,kg^{-1}\,K^{-1}}$, what temperature do you expect the spoon to be at the end of a 2-minute full-power heating cycle?

Solution 2.